Egyptian History AI Agent

Product priorities, agent sequence, tools, and system design

Prepared for the six-member project team | English

Purpose and project status

This handoff consolidates the project's approved product requirements, architecture decisions, agent behavior, and prioritized delivery plan into one working reference for the team. It assumes a greenfield implementation. It contains no schedule estimates and assigns no individual roles; the team will distribute the work itself.

The product is a public, bilingual visitor web app for the Egyptian Grand Museum (GEM), built on React, FastAPI, PostgreSQL with vector search, Redis, and Docker, with a Central Concierge orchestrator. The exact AI, speech, map-data, and hosting providers remain open choices behind replaceable adapters. Section 1 gives the full vision and scope.

Priority order agreed with the product owner: (1) museum map and place discovery, (2) grounded AI assistance, (3) historical-character conversations, (4) self-guided tours, (5) accounts and saved state, and (6) turn-based voice. Arabic and English are required from the first release.

How to use this handoff

Use Sections 1–2 for the product scope and release priorities; Sections 3–5 for the agent flow, system design, and safety controls; Sections 6–7 for verification and work packages; Section 8 for the feature/story inventory; Section 9 for open decisions; and Section 10 for the free-tool comparison. Open map data and approved historical source material are marked as research dependencies.

Stakeholder analysis

Stakeholders are grouped by the value they receive, the decisions they control, and the safeguards they require.

Stakeholder	Interest	Influence	Needs and concerns
Visitors and tourists	High	High	Easy discovery, map/place context, multilingual explanations, accessible voice and text, and confident visit planning.
Museum management	High	High	Accurate public information, visitor engagement, safe AI behavior, analytics, and alignment with official operations.
Curators and historians	High	High	Reviewable sources, content approval workflow, era boundaries, provenance, and correction controls.
Museum and visitor-services staff	High	Medium	Current galleries, facilities, parking, seating, accessibility, closures, and operational updates.

Stakeholder	Interest	Influence	Needs and concerns
Educators and families	Medium	Medium	Age-appropriate explanations, authored tours, learning prompts, and accessible content.
Project team	High	High	Clear contracts, database ownership, testable agent behavior, provider gateways, and deployment documentation.
Content reviewers	High	High	Approval status, source metadata, version history, bilingual review, and safe publication controls.

Stakeholder communication rule: visitor-facing content, locations, and historical answers require an approved source and a named review path. The Concierge can explain or retrieve approved material, but it cannot approve museum content or invent a location.

1 Product vision and scope

Vision

Help tourists discover GEM before a visit and explore it with confidence on site: inspire them with authentic museum and artefact stories, then show what an object or place is, which era it belongs to, where it is, and what is nearby.

Primary audiences

Audience	Need the product serves
Tourists planning a Cairo visit	Understand why GEM is worth visiting, preview highlights and tours, and open official visit information.
Visitors inside GEM	Find a gallery, artefact, parking, garden, seating area, or facility; see nearby places and relevant stories.
Curious visitors and families	Learn about eras, kings, queens, and artefacts through clear explanations and bounded character conversations.

Product boundary

The application is not a replacement for GEM ticketing, payments, or the museum's official visitor operations. The map is a museum-area discovery map, not live positioning or turn-by-turn navigation. Character conversation is educational role-play, not unrestricted general-purpose chat. The public experience must continue to work if AI or voice services fail.

Included in the product

- Public discovery of GEM, its story, nearby attractions, and official visit/ticket handoff.
- A visual museum-area map with places, nearby points of interest, filters, and related content.
- Approved artefact and era discovery in Arabic and English, with clear translation availability.
- A Central Concierge that routes visitor questions to specialist capabilities and returns one coherent answer.

- Khufu, Ahmose I, and Cleopatra VII as historically bounded, clearly labelled character guides.
- Curated self-guided tours; later account-based saving and turn-based push-to-talk STT/TTS.

Explicitly out of scope

- Ticket sales, payments, refunds, or booking workflows.
- Indoor GPS, beacons, QR-based positioning, or turn-by-turn routes.
- Autonomous actions that change account state without visitor confirmation.
- A staff CMS, scholarly catalogue, or public launch claim about unverified place coordinates.

2 Feature priorities and delivery phases

Phases express dependency and priority only; no durations or dates are estimated. Advance a phase when its acceptance evidence is met and its required content or external dependencies are ready.

Priority	Feature group	Release intent	Core outcome
P0 — foundation	Trust, content, platform, bilingual baseline	Required across all phases	Approved content and map data have owners, review states, language metadata, and traceable source records; the app has secure boundaries and safe fallbacks.
P1	Museum discovery and map	First visitor-value release	Visitors can discover GEM and browse a trustworthy museum-area map, select a place, see nearby places, and open related information.
P2	Grounded Concierge and history	First visitor-value release	Visitors can ask museum-scoped questions and explore eras, artefacts, and related stories using approved content.
P3	Historical characters	First visitor-value release	Visitors can choose Khufu, Ahmose I, or Cleopatra VII and receive grounded, era/lifetime-bounded replies.
P4	Self-guided tours	Next priority after core discovery	Visitors can select a curated tour, view ordered stops on the map, and mark progress.
P5	Accounts and saved state	Next priority after core discovery	Visitors can sign up and save favourites, tours, language preference, and conversations; public browsing remains available without sign-in.
P6	Push-to-talk voice	Later priority	Visitors record one question, review its transcript, receive the normal safe text answer, and optionally play speech.

Phase 0 — Establish trusted foundations

Create the modular FastAPI application and React client, Docker/Compose development environment, PostgreSQL schema and migrations, Redis connection, agent contracts, provider gateways, trace identifiers, and test harness. Define approved-content records and their review lifecycle. Research and validate the museum map source, historical source corpus, image rights, and bilingual content coverage before promising precise public facts.

Phase 1 — Make GEM discoverable and explorable

Deliver bilingual public discovery, the museum story, official visit handoff, map browsing, place details with nearby relationships, artefact/era pages, and a grounded Concierge with the Safety and

Provenance Guard on every generated answer. Add the three bounded characters once retrieval and safety pass the evaluation scenarios.

Phase 2 — Build repeatable visits

Add curated tours and progression, then accounts, recovery, favourites, and save/resume for tours and conversations. Language selection stays public from Phase 1; only the language preference is saved with the account.

Phase 3 — Add turn-based voice

Add the push-to-talk voice flow described under “Voice sequence” in Section 3, with Arabic/English STT, transcript review, optional TTS, and text fallbacks.

Release gates

Gate	Evidence required before moving on
Content readiness	Each public record has a source, approval status, language status, owner/reviewer, and update date; missing map facts are shown as unavailable, never guessed.
Agent readiness	Structured contracts validate; no specialist-to-specialist calls; prompt-injection, unsupported-fact, out-of-era, timeout, and outage tests pass.
Visitor readiness	Core journeys work on mobile and desktop in Arabic RTL and English LTR; keyboard and screen-reader paths work; map scope is clear.
Voice readiness	Mic is requested only after an explicit action; transcript is visible; TTS failure preserves text; raw audio is discarded by default.

3 Agent model and request sequence

The visitor experience should feel inviting and story-led, with a strong museum or artefact image and a clear next action on the first screen. Home introduces the museum; Explore and Search help visitors browse; the Map shows a selected place, its area, gallery image, and nearby locations; artefact pages connect approved history to its museum location; and the Concierge offers text or turn-based voice help. Arabic and English, readable mobile layouts, and essential visitor information remain available throughout.

The visitor sees one assistant: the Museum Concierge. Specialist agents are internal capabilities, not separate bots in the user interface. The Concierge alone owns orchestration, selects the required specialists, combines results, and presents the final visitor-facing answer.

Figure 1 makes intent routing and the question-time RAG path explicit, from the visitor's request through evidence retrieval and safety review.

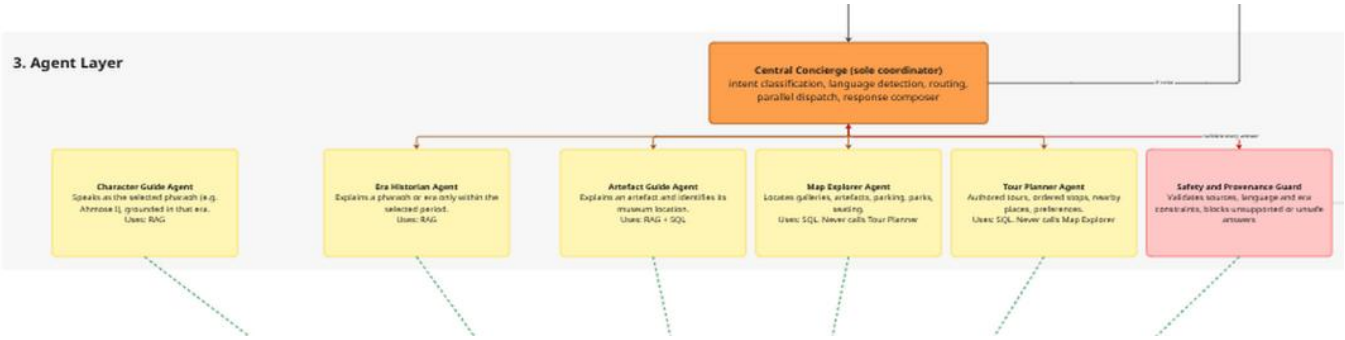


Figure 1. Concierge intent routing, RAG retrieval, and guarded response workflow.

Capability	Responsibility	May access	Must not do
Central Concierge	Classify intent, ask a focused clarification if needed, plan calls, combine results, and propose next actions.	Visitor context, allowed agent contracts, application services.	Bypass safety, invent museum records, or persist changes without authorization/confirmation.
Map Explorer	Resolve a place and return its approved location, nearby places, categories, and related links.	Approved map/place records and map domain port.	Claim GPS certainty, infer missing coordinates, or create route directions.
Era Historian	Explain an artefact, era, person, or relationship using approved content.	Published content retrieval filtered by language, scope, and era.	Turn unsupported retrieval into fact or silently use unapproved content.
Character Guide	Answer in the selected character framing within configured lifetime and era boundaries.	Character configuration and approved knowledge set.	Claim personal knowledge of later events or answer as a character outside its scope.
Tour Planner	Select/adapt an authored tour using visitor interest and available stops.	Published tours, stop order, map/place records.	Invent unavailable stops or persist a tour without visitor confirmation.
Safety and Provenance Guard	Evaluate scope, citations, era boundary, language, privacy, and response safety.	Candidate answer, citations, policies, character constraints.	Be exposed as another conversational persona or allow unsupported claims through.
AI and Voice Gateways	Wrap provider calls with limits, timeouts, policy, retries, and provider adapters.	Approved prompts/text or ephemeral audio as applicable.	Own application state or expose provider secrets to the browser.

Standard text request sequence

Step	Component	Action and handoff
1	Visitor / React	Visitor asks a question or selects a map, artefact, era, character, or tour context. The browser sends locale, context, and request ID to FastAPI.
2	FastAPI boundary	Authenticate if needed, validate the public request, apply rate limits, create trace/conversation context, and call the application service.
3	Central Concierge	Determine intent and create an explicit plan. It calls only the minimum specialist capabilities needed; specialists do not call each other.
4	Specialist capabilities	Retrieve approved content and return structured findings, citations/provenance, uncertainty, and typed proposed actions.

Step	Component	Action and handoff
5	Safety and Provenance Guard	Check museum scope, approved citations, language, character era/lifetime rules, prompt injection, privacy, and response claims. Allow, qualify, redirect, or block.
6	Central Concierge	Combine permitted results into one concise answer, display sources/related content, and offer clear next actions.
7	React / visitor	Render text and citations. Require explicit visitor confirmation for persistent or external actions where specified.
8	Observability	Record request/trace IDs, agent and contract versions, source IDs, latency, usage, safety outcome, and typed errors; redact secrets and raw audio.

Specialist execution rules

- Every internal call uses a versioned JSON contract with request_id, trace_id, conversation_id, actor/permissions, locale, deadline_ms, provenance, status, data, proposed_actions, citations, and typed error.
- The Concierge creates every multi-agent sequence. A specialist uses only its approved tools and domain ports and never calls another specialist; the Safety and Provenance Guard gates every generated answer.
- Visitors see one coherent response. Ask a clarification only when the missing context changes the answer or recommendation.
- Unavailable services produce a truthful fallback: static approved content/map when possible, otherwise a clear limitation and retry path.

Voice sequence

Voice is a transport adapter around the same text path: visitor presses and holds or explicitly starts recording → browser requests microphone access → audio is submitted once → Voice Gateway transcribes → visitor can see/correct the transcript → FastAPI invokes the normal Concierge and safety flow → text answer is returned → TTS may synthesize that exact approved answer → audio is played while text remains visible. If STT or TTS fails, the visitor can continue with text. Do not retain raw audio by default.

4 System design

Architecture style

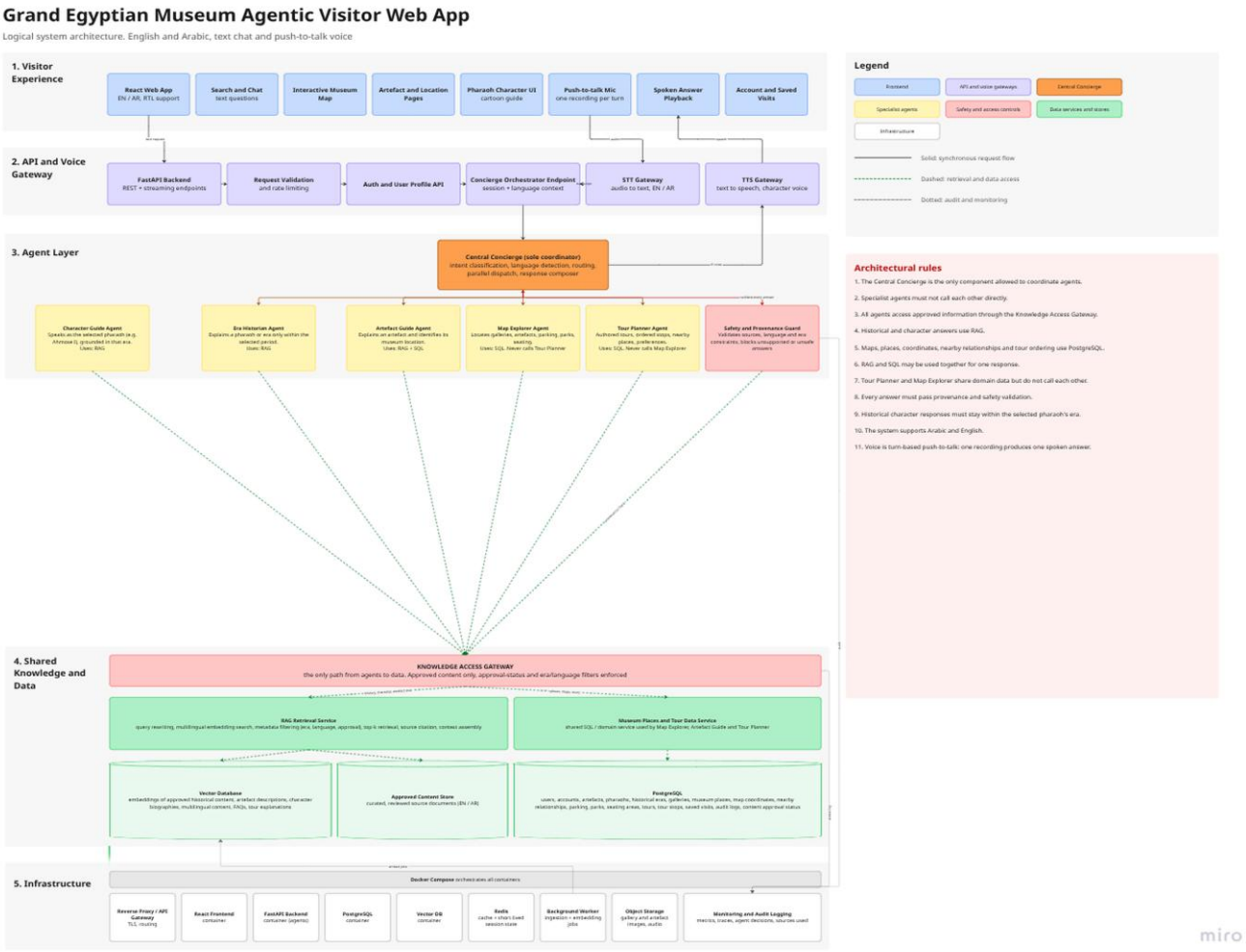
Use a modular monolith for the first release: one FastAPI codebase with clear domain and application boundaries, plus separate web and worker containers. This keeps deployment and local setup manageable for a six-person team while contracts preserve the option to extract a service later if measured scale or ownership requires it.

Layer	Responsibilities	Key rule
React web	Public discovery, map views, content pages, Concierge chat, tour/account/voice controls.	No provider secrets or direct calls to LLM/STT/TTS providers. English LTR and Arabic RTL; text is always available.
FastAPI API	Public endpoints, auth boundary, request validation, application services, OpenAPI contract.	Authorize at API/domain boundaries; do not delegate access control to prompts.

Layer	Responsibilities	Key rule
Application services	Use cases and orchestration; Central Concierge and approved action confirmation.	Concierge is the single owner of multi-agent request planning.
Domain modules	Content, map, tours, accounts/visitor session, conversation, policies, audit.	Each module owns its tables and validates writes.
Ports and adapters	AI Gateway, Voice Gateway, search, database, cache/queue, object storage, external official links.	Provider implementations remain replaceable; enforce timeouts, budgets, retry limits, and redaction.
Worker	Async indexing, media, exports, and other long-running jobs.	Durable/idempotent job requests with status visible to owning module.
Data services	PostgreSQL truth, vector extension/search, Redis ephemeral context and coordination.	Approved content and durable state belong in PostgreSQL; Redis is not the source of truth.

Runtime topology

From	To	Purpose
Visitor browser	React container	Renders the public interface and sends user-initiated requests.
React	FastAPI	Typed HTTPS API; authenticated endpoints only for account-owned data.
FastAPI	Concierge	Validated application context and explicit visitor intent.
Concierge	Map / Era / Character / Tour capabilities	Versioned structured agent contracts; no peer calls.
Specialists	Content/Search ports	Approved database rows, vector retrieval, map relations, and authored tour data.
Concierge	Safety and Provenance Guard	Candidate response plus exact content references and policy context.
Gateways	LLM / STT / TTS adapters	Provider APIs or local models, selected by configuration and policy.
API / Worker	PostgreSQL and Redis	Durable state and approved content in Postgres; ephemeral cache/context/queue coordination in Redis.



Data ownership and core entities

Owner	Entities and records
Content module	ApprovedContent, LanguageVariant, Citation/Source, Era, Character, Artefact, MuseumPlace, nearby relationships, Tour, TourStop, content review state, audit records.
Visitor Session / Account module	Account, authenticated session, locale preference, saved place, saved tour, saved conversation, conversation and turn metadata.
Agent / audit boundary	AgentTurn trace metadata and source references; no raw audio by default; no provider secret or token in logs.

Figure 2 shows both RAG lifecycles: the approved-content indexing pipeline and the filtered retrieval path, alongside the relational/vector database boundary.

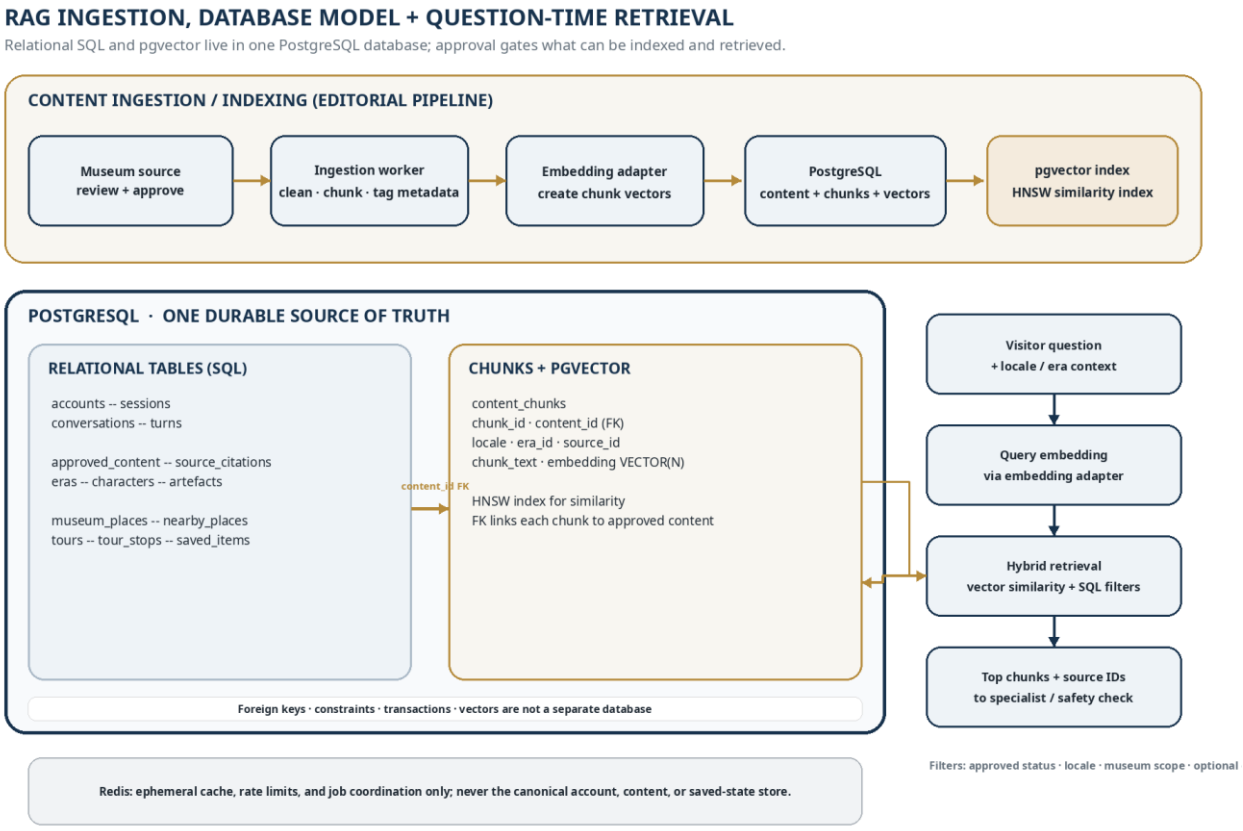


Figure 2. Approved-content ingestion, PostgreSQL SQL tables, pgvector, and RAG retrieval.

Example retrieval query (illustrative SQL; use a parameterized query in the application):

```
SELECT cc.content_id, cc.chunk_text, cc.source_id

FROM content_chunks AS cc

JOIN approved_content AS ac ON ac.id = cc.content_id

WHERE ac.approval_status = 'approved'

AND ac.museum_scope = :museum_scope

AND cc.locale = :locale

AND (:era_id IS NULL OR ac.era_id = :era_id)

ORDER BY cc.embedding <=> :query_embedding

LIMIT 5;
```

The worker creates embeddings only from approved content; retrieval combines SQL filters with vector similarity and returns original content and source IDs for citation and safety review.

Common conventions: UUIDs for public entities and opaque session IDs; UTC timestamps; ISO 8601 serialization; BCP 47 locale values; snake_case in APIs and database fields; stable client-safe error codes; explicit idempotency for retryable writes.

Map data design

Treat the detailed indoor/site map as an unresolved research dependency. The official GEM visitor map is the first source to investigate, but the team must verify whether it contains sufficient location geometry and reuse rights. If no structured map is supplied, create a reviewed digital map dataset from an authorized plan and manual curation. Store each place with category, floor/area, geometry or position reference, neighboring place IDs, accessibility/visitor metadata, source, review status, and language variants. Never infer a precise artefact location from a screenshot or general text.

Suggested project modules

apps/web/src/features/{discovery,map,history,characters,tours,account,voice}; apps/api/app/{api,application,agents,domain,ports,adapters,contracts,observability}; apps/worker/jobs; infra/{docker,migrations,deploy}; tests/{unit,contract,integration,e2e}.

5 Safety, privacy, and provenance

Map Explorer and Tour Planner read the same approved museum-place and tour data service, and historical and character explanations use the shared RAG service. The Central Concierge coordinates any request that needs several capabilities.

The Safety and Provenance Guard is an application-enforced control point before an answer reaches the visitor. Prompts can guide behavior, but approval checks, authorization, privacy, and write confirmation must be enforced by code and data policy.

Risk	Required control	Failure behavior
Hallucinated museum/history facts	Retrieve only approved records, filter by scope/language/era, require citations for factual claims, and capture content IDs in trace data.	Qualify uncertainty, request clarification, or say the answer is unavailable; never fabricate a location or citation.
Character speaks beyond its lifetime/era	Configure explicit lifetime/era boundaries and evaluate responses against character context before display.	Stay in role, state the limit, and redirect to a relevant historian/content page.
Prompt injection or unsafe request	Treat retrieved/user text as untrusted data; apply server-side system policy, retrieval allow-lists, output checks, and red-team cases.	Ignore attempts to override policy; safely refuse or redirect.
Unauthorized state change	Typed proposed actions, domain authorization, idempotency, and explicit visitor confirmation for save/adapt/external actions.	Do not commit the mutation until authorized and confirmed.
Sensitive audio/transcript	Request mic only after user action; ephemeral processing; raw audio discarded after the turn; clear retention/consent settings for saved conversations.	Text fallback on errors; do not log raw audio, secrets, or private conversation content unnecessarily.
Provider or agent outage	Timeouts, bounded retries, rate limits, fallback adapters, health checks, tracing, and static content/map routes.	Show a clear limitation and preserve current visitor context.
Incorrect or outdated published record	Every record has a source, owner, review status, language status, and updated timestamp; correction/unpublish path.	Withdraw or mark content unavailable until re-approved.

Safety response outcomes

- Allow: answer is in scope and supported by approved sources.

- Qualify: answer is mostly supported but needs a visible uncertainty note.
- Clarify: one missing detail materially changes the result.
- Redirect: question is beyond a character's era or the product's scope; offer an approved relevant path.
- Block: unsupported, unsafe, privacy-invasive, or unauthorized request; provide a concise explanation and safe alternative.

6 Nonfunctional requirements and verification

Area	Requirement to verify
Accessibility	Core journeys target WCAG 2.2 AA-equivalent practices: keyboard, focus, contrast, semantic structure, alternative text, and screen-reader labels.
Bilingual experience	Arabic RTL and English LTR layouts; explicit translation availability and approved fallback; no silent machine translation presented as reviewed fact.
Performance	Set quantitative budgets during implementation planning for common mobile networks. Keep map and public approved content usable without AI.
Reliability	AI, map, and voice failures expose loading, timeout, retry, and safe fallback states; core public content remains available during AI outages.
Privacy/security	Collect only necessary data; protect credentials and saved content; explain consent and retention; do not train/expose private content without explicit policy/consent.
Content operations	Owners can review, correct, or unpublish records without a full app release; every record has owner/reviewer, source, language, status, and timestamp.
Observability	Structured logs and distributed traces capture request IDs, agent/provider spans, citations, usage, latency, safety outcome, and typed error without secrets/raw audio.
Deployment	API, web, and worker build as reproducible Docker images; Compose supports local development; config uses environment variables and health checks.

Minimum evaluation scenarios

- A map lookup names an approved place and nearby points; it never promises turn-by-turn directions or claims an unverified position.
- A character refuses or redirects a question about events outside the configured lifetime/era, while remaining clearly labelled as role-play.
- An unsupported historical claim or prompt injection does not bypass approved retrieval or safety policy.
- English and Arabic requests preserve the same intent when approved translations exist; RTL display and fallback language are correct.
- A one-hour tour request returns only a valid authored/approved sequence and explains the reason for each included stop.
- STT failure returns to typed input; TTS failure preserves the answer text; raw audio is absent from persisted logs/storage.
- Visitor-confirmed save/resume is isolated by account authorization; anonymous visitors can still use public map and discovery content.

7 Team work packages

The following work packages describe outcomes and dependencies only; the six team members can divide, pair, or rotate them as they prefer.

Work package	Outcome	Depends on / connects to
Product and content model	Approved record schema, bilingual content inventory, editorial review rules, source/licensing register.	Museum content/source research; required before publishing factual AI responses.
Map investigation and dataset	Verified source plan, place taxonomy, area/floor model, coordinate/geometry format, nearby relationships, known coverage gaps.	Official GEM map access/reuse rights; map UI must display completeness status.
Web experience	Discovery, map, content, Concierge, account/tour/voice views with responsive and bilingual directionality.	API contracts and public content states.
API and domain services	FastAPI endpoints, account/session boundaries, content/map/tour/conversation use cases, validation, auth, migrations.	Domain ownership and versioned public API schemas.
Agent orchestration and retrieval	Concierge plan/dispatch, specialist contracts, approved retrieval, citations, guardrails, evaluation cases.	Approved content dataset and safety policy.
Infrastructure and delivery	Docker/Compose, database/Redis, worker, environment examples, health checks, CI, deployment investigation.	Hosting decision and security review before public release.
Verification and release readiness	Contract, integration, E2E, accessibility, bilingual, safety, map accuracy, and voice fallback tests.	All feature packages; release gates in Section 2.

Working agreements

- Agree on agent JSON contracts and data ownership before parallel implementation.
- Use feature flags for optional provider integrations and retain a deterministic static fallback.
- Review the prioritized backlog together when map/source discoveries change scope; update the PRD, architecture, and stories consistently.

8 Feature and story inventory

This inventory summarizes the approved epics and stories; detailed acceptance criteria remain in the project's BMAD epics source file. Priorities follow the product owner's feature order.

Priority	Epic	Stories
P1	Epic 1 Discover GEM and plan a visit	1.1 Discover GEM from the home page; 1.2 Read the museum story; 1.3 Open official visit information.
P1	Epic 2 Explore the museum-area map	2.1 Browse and filter the map; 2.2 View a place and nearby points; 2.3 Clarify map scope and no turn-by-turn navigation.
P2	Epic 3 Explore artefacts, eras, and related stories	3.1 Browse/search artefacts; 3.2 View artefact details; 3.3 Explore historical eras; 3.4 Follow related content.
P3	Epic 4 Learn with historical characters	4.1 Choose Khufu, Ahmose I, or Cleopatra VII; 4.2 Receive a grounded, era/lifetime-bounded answer with source context.
P4	Epic 5 Follow self-guided tours	5.1 Browse authored tours; 5.2 Follow ordered stops and mark progress.
P5	Epic 6 Save and personalize a visit	6.1 Create an account and sign in; 6.2 Recover account access; 6.3 Manage saved favourites; 6.4 Choose a language during exploration; 6.5 Save language preference and show translation availability; 6.6 Use accessible core flows; 6.7 Save/resume a character conversation; 6.8 Save/resume a tour.
P6	Epic 7 Ask and hear a voice question	7.1 Grant mic access and record/cancel/submit; 7.2 Transcribe through the normal safe Concierge path; 7.3 Play/pause/replay TTS with text fallback; 7.4 Discard raw audio by default and apply transcript controls.

Cross-cutting requirements

Approval and provenance, Arabic/English integrity, accessibility, security, privacy, observability, and failure fallback apply to every epic from the first implementation slice.

Traceability summary

The PRD contains 26 functional requirements (FR-1 through FR-26). The current epic breakdown maps every requirement to seven epics and 25 user stories. For implementation, keep story-level FR tags and acceptance criteria in the source backlog so test coverage remains traceable.

9 Open research and decisions

Open item	Current state	Decision evidence needed

Open item	Current state	Decision evidence needed
Detailed GEM map source	Open research; the team will investigate.	Official digital map/source plan, floor/area completeness, format, coordinate reference, updates, accessibility details, reuse rights, and validation contact.
Approved historical corpus	Open research; the team will investigate.	Museum-published sources and any supplemental reputable materials; reviewer and approval process; source/citation granularity; translations and image rights.
Exact LLM/STT/TTS providers	Not selected; compare no-cost local and cloud free-tier options in the final section.	Arabic Egyptian and English quality, privacy/data use, rate limits, latency, Docker deployment, API stability, operational cost after quotas, provider terms.
Production hosting	Cloud-neutral Docker architecture; host not selected.	Expected public traffic, data residency, database/Redis cost, backups, observability, availability, and team operations.
Identity provider	Account boundaries defined; vendor/protocol remains deferred.	Email verification/recovery, session security, privacy obligations, deployment effort, and provider limits.
Editorial/CMS workflow	Approved-content lifecycle required; implementation is open.	Who reviews and publishes records, correction/unpublish workflow, import format, permissions, audit requirements.
Voice character/accent	Arabic and English voice needed; exact voice not selected.	Listening tests with Egyptian Arabic and visitor-appropriate English; clarity, dialect fit, cultural review, licensing, and visitor consent.

Recommended next decisions

1. Request the most detailed GEM map file or permission pathway and record what each map layer actually covers.
2. Create a small bilingual content pack with reviewed sources for one gallery/area, a few artefacts, and the three character boundaries.
3. Run short Arabic/English evaluation tests across candidate models before wiring a production provider.
4. Approve a retention/consent policy for saved chat and voice transcripts before account and voice release.
5. Keep the first map implementation data-driven so corrected coordinates or an updated official plan do not require UI rewrites.

10 Free tools and provider comparison

The comparison below separates truly self-hosted/open-source components from cloud free tiers. “Free” does not mean production-ready at public scale: local tools require hardware and operations; cloud free tiers impose quotas, terms, and data-handling conditions. Verify terms and Arabic/English performance again before production selection. Provider comparisons were checked on 10 October 2026.

LLM options

Option	Free model	Strengths	Limits and risks	Best fit

Option	Free model	Strengths	Limits and risks	Best fit
Ollama + a locally licensed model	No per-token API bill; local execution.	Keeps prompts and visitor content on controlled hardware; replaceable local API; works in Docker-based dev.	Needs capable CPU/GPU/RAM; model license varies; throughput and Arabic quality must be tested; local compute and hosting are not cost-free.	Privacy-first development, evaluation, or a self-hosted deployment where hardware is available.
Google Gemini API free tier	Free tier currently listed for selected models; limits vary by model/project.	Fast start; capable general models; Arabic and English; official API and current model docs.	Quota/rate limits can change; Google lists free-tier content as potentially used to improve products; not a default for private production conversations without policy review.	Prototype and controlled evaluation using non-sensitive test data.
Groq API free plan	Free plan with model-specific request/token limits.	Fast inference; multilingual candidate models; OpenAI-compatible API style; free plan has published caps.	Hard limits and 429s; model availability changes; third-party processing; verify retention/privacy terms and test Arabic GEM terminology.	Prototype or classroom demonstration with quota fallback and no sensitive data.

Recommendation: build the AI Gateway first and benchmark two candidates against the same approved-content evaluation set. Keep provider selection configurable. For public launch, choose only after reviewing data use, privacy terms, rate limits, and projected spend; do not let a prototype free tier become an accidental production dependency.

Speech-to-text options

Option	Cost model	Arabic and English	Trade-offs
whisper.cpp	Open-source local inference; no per-minute API fee.	Whisper is multilingual; verify Egyptian Arabic and noisy-gallery performance with recorded test clips.	Requires local CPU/GPU and model download; latency and memory depend on model/hardware; useful for privacy and offline fallback.
Groq Whisper API	Cloud free plan subject to published audio-seconds/request quotas.	Whisper Large v3 / Turbo are multilingual.	Simple API; hard quota and external processing; free plan can throttle; add typed fallback and do not store raw audio by default.
Gemini 3.5 Transcribe	Cloud API free tier subject to changing model quotas.	Official support list includes Arabic (Egypt, ar-EG); test dialect/code-switching.	Strong language hints and transcription workflow; free-tier data-use terms and quotas need review; cloud audio leaves app infrastructure.

Text-to-speech options

Option	Cost model	Language fit	Trade-offs
Browser SpeechSynthesis	No application API bill; uses voices available on visitor device/browser.	Arabic/English availability and voice quality vary by browser/OS.	Fast MVP fallback; inconsistent voice selection/pronunciation; not a controlled character voice and may not work uniformly.

Option	Cost model	Language fit	Trade-offs
Groq Orpheus TTS	Free-plan quota/limits apply; verify current account limits.	English and Arabic Saudi dialect model are listed.	Expressive and easy to test; Arabic is Saudi, not Egyptian; current docs limit input length to 200 characters, so long answers require chunking; cloud data processing.
Gemini TTS free tier	Pricing currently lists free input/output for selected TTS model; model limits apply.	Official docs list broad multilingual support; test Egyptian Arabic and English voice quality.	Promising quality and controls; free tier states content may be used to improve products; no TTS streaming for current recommended flow; requires privacy review.
Piper local TTS	No per-request API bill; self-hosted compute.	A listed Arabic voice is Jordanian; English voices also exist.	Check voice/model card licensing individually; current maintained Piper project is GPL-3.0 and intended for personal/research use, so obtain licensing review before incorporating in a public/commercial app.

Recommendation: retain visible text as the canonical answer. For early testing, compare browser speech, Groq Orpheus, and Gemini TTS using museum-reviewed Arabic and English samples. Do not label a Saudi Arabic voice as Egyptian. Review licenses before shipping local voice models.

Map and vector tools

Area	Free/open option	Comparison and caveat
Map rendering	Leaflet with a replaceable tile provider; custom GEM floor/site-plan layer.	Leaflet is an open-source interactive-map library. It does not supply a verified GEM indoor map. OpenStreetMap data has attribution requirements, and its community tile servers are best-effort with strict usage/caching rules; for sustained public traffic, choose an allowed provider or self-host tiles.
Vector retrieval	pgvector inside PostgreSQL; Qdrant as an alternative if independent vector operations are later justified.	pgvector best matches the agreed PostgreSQL-first design and keeps structured content and vectors together. Qdrant is also open source and can run locally, but adds another service and operational boundary. Start with pgvector unless measured search scale or features justify separation.
General platform	React, FastAPI, PostgreSQL, Docker Compose.	Already selected in project architecture. These are software components, not zero-cost production hosting: production compute, storage, backups, domains, monitoring, and provider quotas can incur charges.

Comparison method for team evaluation

Use one small repeatable test pack: 20–30 museum questions in Arabic (including Egyptian colloquial phrasing) and English; approved answer and source for each; 10 out-of-era character prompts; 5 prompt-injection/unsupported-fact prompts; and short quiet/noisy audio clips. Score factual grounding, correct refusal/qualification, source coverage, language quality, latency, failure behavior, privacy terms, setup effort, and cost when quotas are exceeded. The scorecard should decide the provider; this handoff does not claim that any free tier is approved for public production.

Primary references

Gemini API pricing and free-tier data-use notes — https://ai.google.dev/gemini-api/docs/pricing. Includes current model-level free-tier and pricing information.

Gemini API usage tiers — https://ai.google.dev/gemini-api/docs/rate-limits. Exact rate limits vary by account and model.

Groq API rate limits — https://console.groq.com/docs/rate-limits. Free-plan limits are listed per model and may change.

Groq speech-to-text — https://console.groq.com/docs/speech-to-text. Multilingual Whisper API models.

Gemini transcription and supported languages — https://ai.google.dev/gemini-api/docs/transcribe. Includes Arabic Egypt (ar-EG) in the supported list.

Groq text-to-speech — https://console.groq.com/docs/text-to-speech. Lists English and Arabic Saudi Orpheus models.

Gemini TTS pricing — https://ai.google.dev/gemini-api/docs/pricing. Free-tier terms and rates vary by model and may change.

Gemini speech generation guide — https://ai.google.dev/gemini-api/docs/speech-generation. Model behavior, language support, and streaming limitations.

Whisper.cpp — https://github.com/ggml-org/whisper.cpp. Local Whisper inference implementation and supported platforms.

Piper voice list — https://github.com/OHF-Voice/piper1-gpl/blob/main/docs/VOICES.md. Lists Arabic Jordan voice; review each model card and GPL-3.0 implications.

Web Speech SpeechSynthesis — https://developer.mozilla.org/en-US/docs/Web/API/SpeechSynthesis. Browser/device-provided synthesis voices and controls.

Leaflet documentation — https://leafletjs.com/reference-2.0.0.html. Open-source interactive map library.

OpenStreetMap tile usage policy — https://operations.osmfoundation.org/policies/tiles/. Attribution, caching, availability, and limits for community tile servers.

pgvector — https://github.com/pgvector/pgvector. Open-source vector similarity search for PostgreSQL.

Qdrant local quickstart — https://qdrant.tech/documentation/quickstart/. Self-hosted/local vector database option; adds a separate service.

Ollama model library — https://ollama.com/library. Local model runtime; review each downloaded model's separate license.